Bicol University Polangui
Polangui Albay

Bachelor of Science in Information Systems
IS ELEC 2 - WEB SYSTEMS

FINAL PROJECT - PHASE 1:COMPREHENSIVE PLANNING & ARCHITECTURE DESIGN

Project Title:EVENT-UALLY DONE
Group Name: Event Handlers

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I. PROJECT PROPOSAL

INTRODUCTION

Organizing events requires proper planning and management of materials and equipment, such as tables, chairs, projectors, and sound systems. However, manual inventory methods can lead to inaccurate records, misplaced information, and confusion about item availability. To address these challenges, our group developed EVENT-UALLY DONE, a web-based event inventory management system that helps users organize, monitor, and manage event materials efficiently through a centralized and user-friendly platform.

SYSTEM PURPOSE

The primary purpose of EVENT-UALLY DONE is to provide an organized platform for managing event materials and equipment. It allows users to monitor item quantities and check resource availability. The system aims to reduce recording errors and save time compared to manual inventory management. It also helps users identify items that need restocking before an event. Overall, it supports better planning and efficient resource management.

TARGET USERS

1. **Administrators -** Manages user accounts, school events, inventory records, and system information.
2. **Event Organizer/Staff -** Creates event announcements, updates schedules, requests supplies, and monitors event requirements.
3. **Students -** Views upcoming events, reads announcements, and registers for activities when registration is enabled.

KEY FEATURES

1. **Add Inventory Records –** Allows users to enter new materials and equipment into the system, including relevant item details and quantities.
2. **View Inventory Records –** Enables users to review the list of recorded materials and equipment in one centralized platform.
3. **Update Inventory Information –** Allows users to modify item details and quantities whenever changes occur, helping keep inventory records accurate and up to date.
4. **Delete Inventory Records –** Enables users to remove inventory entries that are no longer needed or are no longer part of the inventory.
5. **Quantity Monitoring –** Helps users keep track of the number of available materials and equipment to support proper resource management.
6. **Item Availability Checking –** Allows users to determine which materials are available for use when preparing for upcoming events.
7. **User-Friendly Interface –** Provides a simple and organized layout that helps users navigate the system and manage inventory information more conveniently.

I. USE CASE DIAGRAM

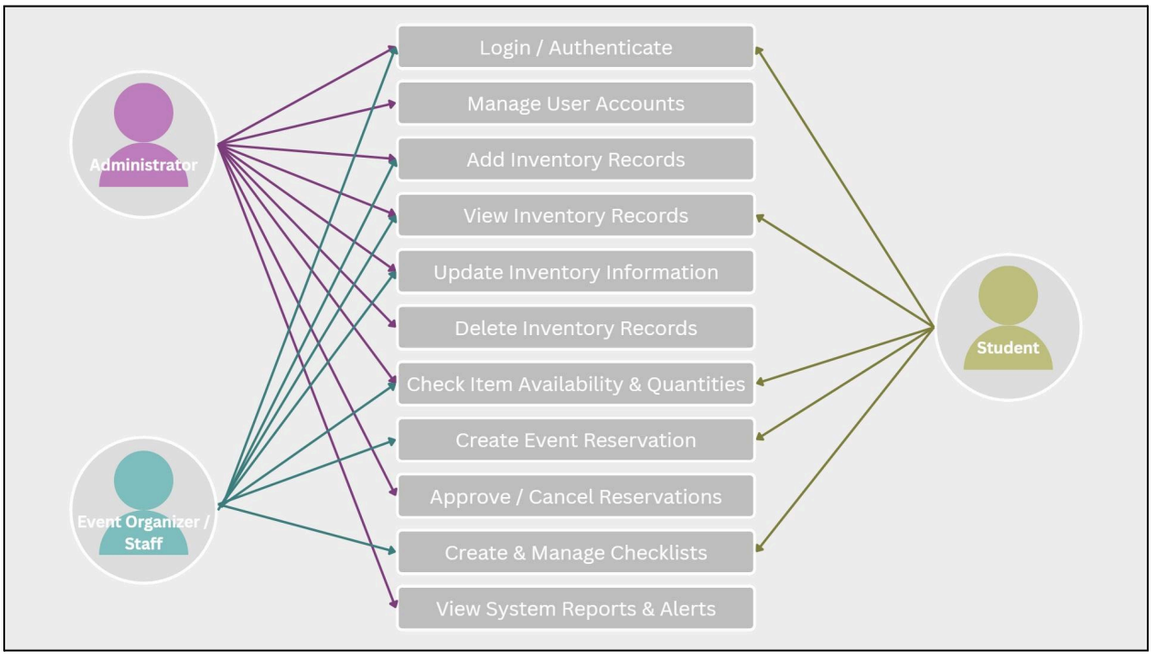


Figure 1: Use Case Diagram

The Use Case Diagram of EVENT-UALLY DONE illustrates the interactions between system users (actors) and the main functional capabilities (use cases) provided by the web-based inventory management system. It defines the system boundary and clarifies how different roles perform specific operational tasks.

Users

1. Administrator

The primary system manager with full oversight and control over the platform. The Administrator performs the following actions:

- **Manage User Accounts:** Creates, updates, and manages account profiles for system users.
- **Manage Inventory Records:** Adds, updates, and deletes equipment entries and monitors stock levels.
- **Approve / Cancel Reservations:** Reviews pending resource requests submitted by users and updates their status.
- **View Reports & Stock Alerts:** Accesses system inventory reports and receives notifications on stock status.

2. Event Organizer / Staff

The primary operational users responsible for planning and executing events. Event Organizers and Staff perform the following actions:

- **View Inventory & Check Availability:** Browses equipment records, checks available quantities, and verifies resource readiness for upcoming events.
- **Add & Update Inventory Information:** Enters new item details and updates item quantities as inventory changes occur.
- **Create Event Reservation:** Books required materials and equipment for specific event dates.
- **Create & Manage Checklists:** Builds preparation checklists to track required event materials and completion progress.

3. Students

General end-users who request equipment for student-led activities and campus events. Students perform the following actions:

- **View Inventory Records:** Browses the centralized inventory list to view available equipment.
- **Check Item Availability:** Verifies if specific materials are available for preparation.
- **Create Event Reservation:** Submits booking requests for materials needed for student activities.
- **Utilize Event Checklists:** Accesses and uses checklists to organize event preparations.

II. ENTITY-RELATIONSHIP DIAGRAM (ERD) - MongoDB Style

The Entity Relationship Diagram (ERD) of EVENT-UALLY DONE illustrates the proposed structure of the database using MongoDB, a NoSQL document-oriented database. Instead of traditional relational tables, MongoDB organizes data into collections containing documents.

The proposed database consists of four main collections: Users, Items, Reservations, and Checklists. These collections organize user account information, inventory records, item reservations, and event preparation checklists. The diagram shows how these collections are connected to support inventory management, quantity monitoring, and item availability checking.

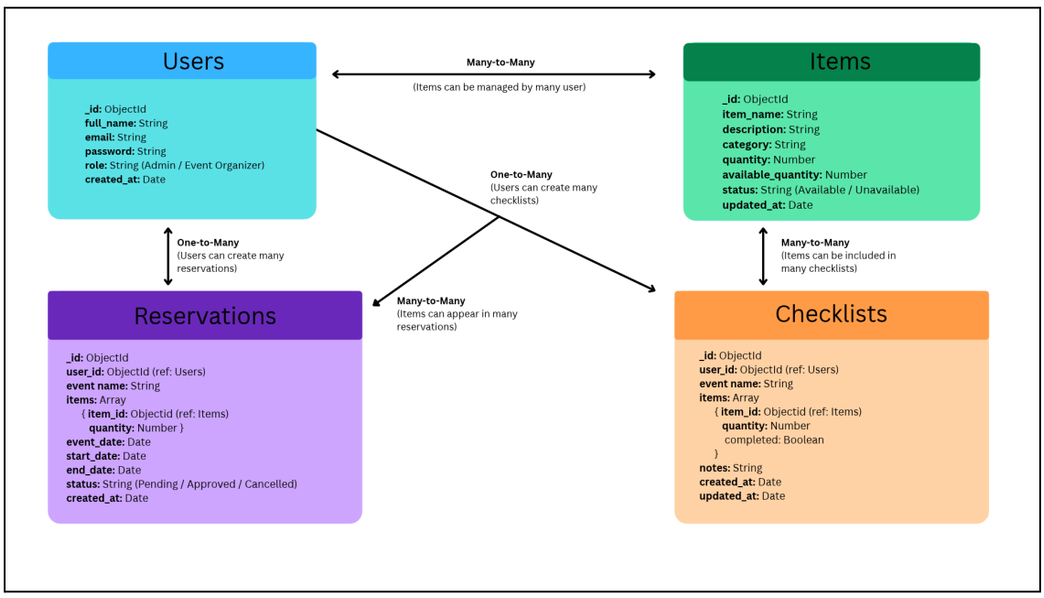


Figure 2: Entity-Relationship Diagram (ERD) of the Event Management System.

Collections and Fields

1. Users Collection

The Users collection stores account information for individuals who access the system, including administrators and event organizers or staff.

Fields:

_id – Unique identifier of the user.
full_name – Full name of the user.
email – User's email address.
password – Hashed password used for authentication.
role – User's assigned role, such as Admin or Event Organizer.
created_at – Date and time the account was created.

2. Items Collection

The Items collection stores information about the materials and equipment managed by the system. It helps users monitor item quantities and availability.

Fields:

_id – Unique identifier of the inventory item.
item_name – Name of the material or equipment.
description – Additional information about the item.
category – Classification of the item.
quantity – Total quantity recorded in the inventory.
available_quantity – Quantity currently available for use or reservation.
status – Current availability status of the item.
updated_at – Date and time the inventory record was last updated.

3. Reservations Collection

The Reservations collection stores records of inventory items reserved for events. It helps organize item requests and monitor the scheduled use of materials and equipment.

Fields:

_id – Unique identifier of the reservation.
user_id – Reference to the user who created the reservation.
event_name – Name of the event associated with the reservation.
items – List of reserved items, including item references and quantities.
event_date – Scheduled date of the event.
start_date – Beginning of the reservation period.
end_date – End of the reservation period.
status – Current reservation status, such as Pending, Approved, or Cancelled.
created_at – Date and time the reservation was created.

4. Checklists Collection

The Checklists collection stores lists of materials and equipment needed for event preparation. It helps users organize required items and track whether checklist tasks have been completed.

Fields:

_id – Unique identifier of the checklist.
user_id – Reference to the user who created the checklist.
event_name – Name of the event associated with the checklist.
items – List of required materials, including item references, quantities, and completion statuses.
notes – Additional reminders or instructions.
created_at – Date and time the checklist was created.
updated_at – Date and time the checklist was last updated.

Relationships Between Collections

The proposed database uses references and embedded documents to connect related information across collections.

1. **Users and Reservations:** One user can create multiple reservations. Each reservation references its creator through the user_id field.
2. **Users and Checklists:** One user can create multiple checklists. Each checklist references its creator through the user_id field.
3. **Items and Reservations:** An inventory item can appear in multiple reservation records over time. The items array stores references to inventory items and their reserved quantities.
4. **Items and Checklists:** Inventory items can be included in multiple checklists. The items array stores item references, quantities, and completion statuses.

III. DATA FLOW DIAGRAM

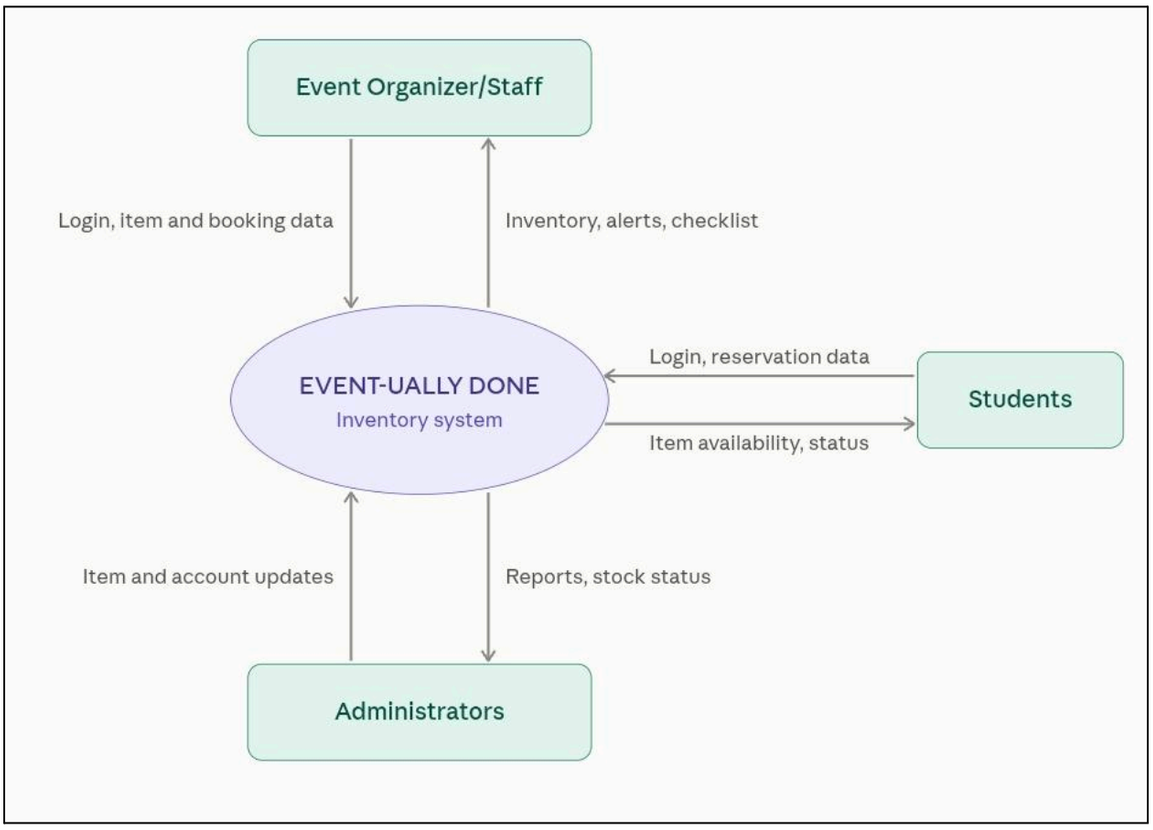


Figure 3. Context Diagram (Level 0)

The context diagram shows EVENT-UALLY DONE as a single system that interacts with three external entities: the Event Organizer/Staff, the Administrators, and the Students. The Event Organizer/Staff send login, item, and booking data to the system, and receive inventory information, alerts, and checklists. The Administrators send item and account updates, and receive reports and stock status. The Students send login and reservation data, and receive item availability and reservation status. This diagram shows the boundary of the system and the main data exchanged with its users.erts, and checklists in return. The Admin sends item and account updates to the system, and receives reports and stock status. This diagram shows the boundary of the system and the main data exchanged with its users.

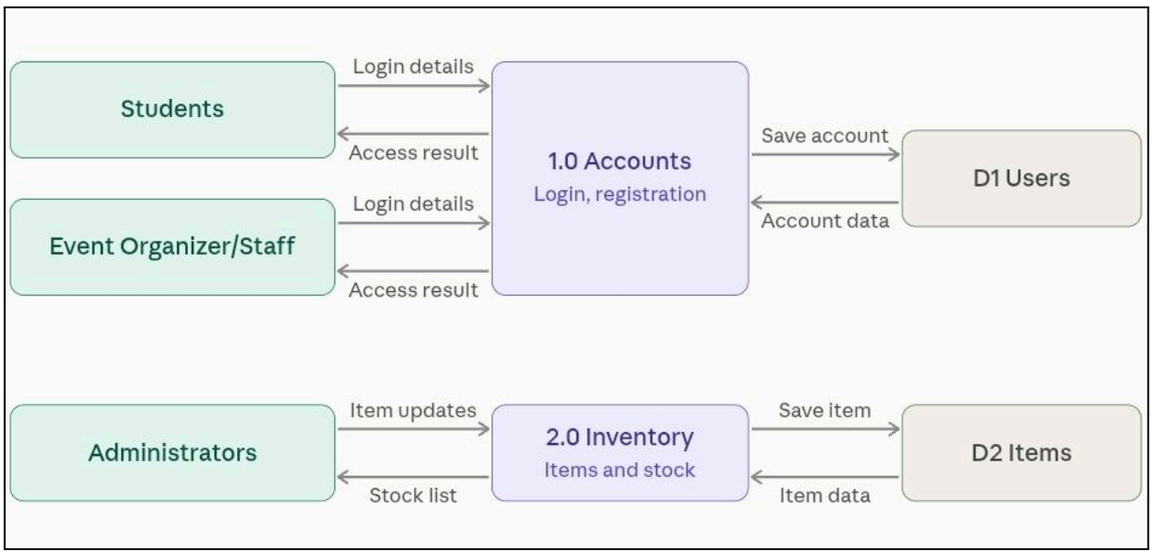


Figure 4. Level 1 DFD: Accounts and Inventory

This diagram breaks down the first two processes of the system. In Process 1.0 (Accounts), the Students and the Event Organizer/Staff submit their login details. The system checks them

against the records in D1 Users and returns the access result. In Process 2.0 (Inventory), the Administrators add, edit, and update items and their stock. The system saves this data in D2 Items and returns the updated stock list.

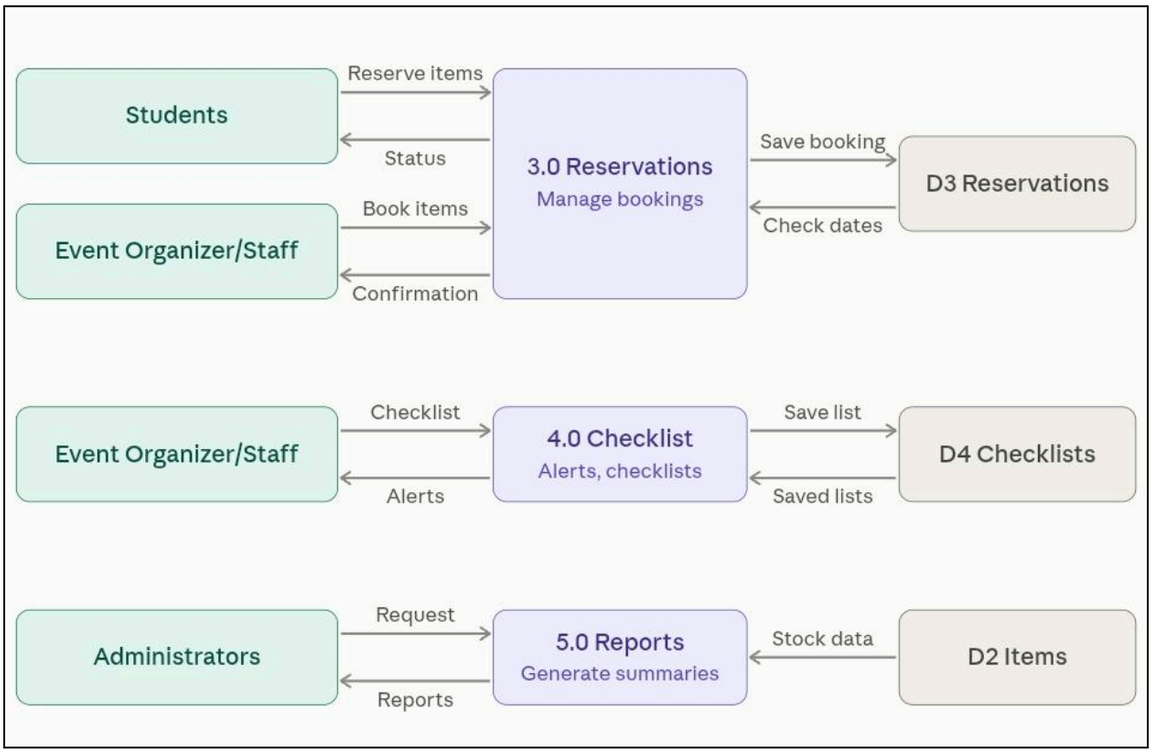


Figure 5. Level 1 DFD: Reservation, Checklist, Reports

This diagram covers the remaining three processes. In Process 3.0 (Reservations), the Students reserve items, and the Event Organizer/Staff book and manage items for events. The system checks the dates against D3 Reservations to avoid double booking, saves the booking, and returns the status or confirmation. In Process 4.0 (Checklist), the Event Organizer/Staff create an event checklist, which is saved in D4 Checklists, and the system returns alerts such as low-stock notices. In Process 5.0 (Reports), the Administrators request a report, and the system gathers stock data from D2 Items to produce a summary.

IV. GitHub Repository Setup

GitHub Repository URL:
https://github.com/gianemarielsanorjo/final-project-Event-uallyDone-BSIS2a

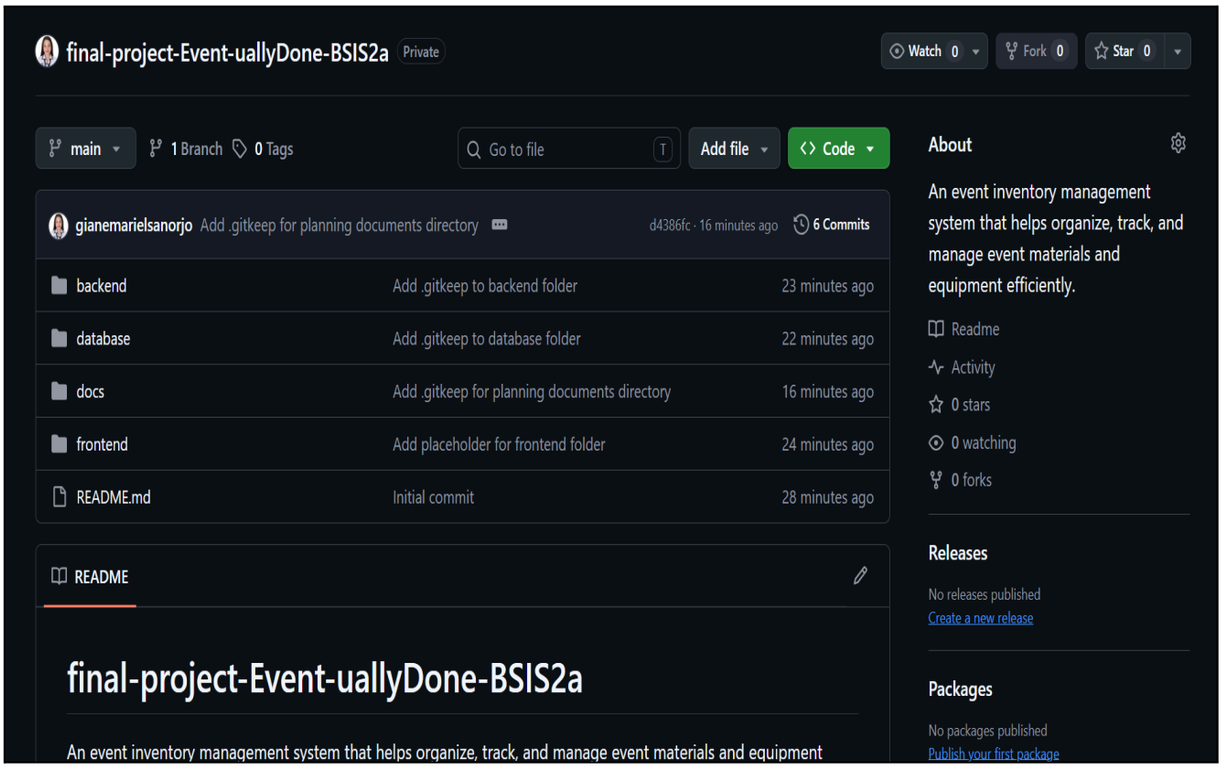


Figure 6: GitHub Repository Structure of the Event-ually Done Project.

V. Personal Role Declaration

OZWALD N. HILADO *(Front-End and Back-End Developer)*

What is your assigned role?

My assigned role is Front-End and Back-End Developer for our final project, EVENT-UALLY DONE. As the Front-End Developer, I am responsible for helping design and develop the website's user interface, including its layout and overall appearance, to make it user-friendly and easy to navigate. As the Back-End Developer, I will help implement the system's functions, handle inventory data, and connect the application to the database to ensure that its features work properly.

Aside from my development role, I was also assigned to create the Entity Relationship Diagram (ERD) using the MongoDB style for our project proposal. My task was to identify and organize the proposed database collections, including Users, Items, Reservations, and Checklists, along with their fields and relationships. This helped present how the system's data would be structured and how the database would support the proposed features of EVENT-UALLY DONE.

What do you commit for changes?

I commit to fulfilling my responsibilities as the Front-End and Back-End Developer by helping develop a functional, organized, and user-friendly website for EVENT-UALLY DONE. I will do my best to create a clear and consistent user interface, implement the necessary system functions, and ensure that the front end, back end, and database work together properly. I also commit to improving my understanding of web development and database management, especially in organizing data and establishing the necessary relationships between collections.

Furthermore, I will communicate with my groupmates to ensure that my work aligns with their assigned tasks and contributions. I will remain open to suggestions, help address technical problems when they arise, and make the necessary improvements based on our project's requirements. Through teamwork, responsibility, and continuous learning, I aim to contribute to the successful development of EVENT-UALLY DONE and help ensure that our final project meets our instructor's expectations.

TINA R. MILLABAS *(Database Manager)*

What is your assigned role?

My assigned role is Database Manager for our final project, EVENT-UALLY DONE. I am responsible for designing and organizing the database that stores the system's information, such as users, items, reservations, and event checklists. I plan the tables, set the primary and foreign keys that connect related data, and make sure the structure is clear enough for our back-end and front-end members to build on. I also make sure the data stays accurate by avoiding duplicate records and double bookings. In addition, I was assigned to prepare the Data Flow Diagram, which shows how data moves between the users (Administrators, Event Organizer/Staff, and Students), the system's processes, and the data stores.

What do you commit for changes?

I commit to keeping our data base organized, accurate, ready for any changes our group needs. When new features or revisions are come up I will update the tables and relationship carefully so they still match the work of our back-end and front-end member. I will tell my groupmates about any changes to the database and the diagrams so that everyone stayed updated, and I will fix the data structure as soon as they are found. And most importantly I will cooperate with my groupmates.

EDGAR ANTER E. RAÑESES *(GitHub Manager)*

What is your assigned role?

My assigned role is GitHub Manager for our final project, EVENT-UALLY DONE. Aside from managing GitHub, I was assigned to create the Use Case Diagram and its descriptions for our project proposal, where I identified our system actors: Administrators, Event Organizers/Staff, and Students, and defined their interactions with key features such as inventory management, availability checking, reservations, and checklists. Being assigned to these tasks taught me the value of structured version control and system design, showing me how proper repository management keeps a team synchronized and how user roles shape system functionality.

What do you commit for changes?

I commit to keeping our GitHub repository clean, updated, and well-organized throughout the project. I will help coordinate code uploads, manage branches, and assist my groupmates in reviewing and merging their work properly. If anyone runs into git errors, merge conflicts, or trouble pushing their files, I will help them troubleshoot and fix it right away. I will also make sure all updated project files and documentation are safely managed so our group always has access to the latest version of our work.

GIANE MARIEL A. SANORJO *(Project Manager & Github Manager)*

What is your assigned role?

My assigned roles are Project Manager and GitHub Manager for our final project, EVENT-UALLY DONE. As the Project Manager, I am responsible for assigning tasks to each group member based on their roles and responsibilities. I help organize our work, explain what needs to be done, and make sure that everyone understands their assigned tasks. I also monitor our progress, check which tasks have been completed, and remind my groupmates about our deadlines. As the GitHub Manager, I am responsible for managing our GitHub repository, organizing our project files and folders, and making sure that the necessary files and updates are properly uploaded. I also help keep our project organized so that our group can easily access and review our work.

What do you commit for changes?

I commit to being responsible and active in leading our group throughout the development of EVENT-UALLY DONE. I will do my best to assign tasks fairly, guide my groupmates when they have questions, and make sure that everyone contributes to the project. I will regularly check our progress, follow up on unfinished tasks, and remind members of important deadlines so that we can avoid delays. I also commit to keeping our GitHub repository updated and organized to ensure that our files are properly managed and accessible to the group. If problems or misunderstandings arise, I will communicate with my groupmates and help find solutions. Most importantly, I will encourage teamwork, cooperation, and open communication so that we can complete our project successfully and submit quality work that meets our instructor's requirements.

MARIA JAMAICA S. VIBAR *(Tester and Debugger)*

What is your assigned role?

My assigned role in our final project, EVENT-UALLY DONE, is Tester and Debugger. As the Tester and Debugger, I will test the website to ensure that its features work properly, including adding, editing, deleting, and viewing event materials. I will also check if inventory quantities are displayed correctly, identify errors, and report them to my groupmates. After the errors are corrected, I will retest the system to make sure the features work as expected.

What do you commit to change?

I commit to becoming more responsible, patient, and confident in performing my assigned tasks. I will carefully test the website, report any errors I find, and retest the system after corrections are made. Through this experience, I aim to improve my

technical skills, communicate effectively with my groupmates, and become more open to feedback and suggestions. I will do my best to contribute to our team's success and help ensure that our final project meets its requirements.